

6(a)	A <i>photon</i> is a quantum of electromagnetic radiation with energy given by $E = hf$ where h is the Plank's constant and f is the frequency of the wave. .
6(b)(i)	$E = hf = \frac{hc}{\lambda}$ $E = 6.63 \times 10^{-34} \times \frac{3 \times 10^8}{680 \times 10^{-9}} = 2.93 \times 10^{-19} \text{ J}$

6(b)(ii)	$\text{Intensity} = \frac{\text{Power}}{\text{Area}} = \frac{P}{A}$ $I = \left(\frac{N}{t} \right) \frac{E}{A}$ $\left(\frac{N}{t} \right) = \frac{I}{E} A$ $= \frac{(3.1 \times 10^3)}{2.93 \times 10^{-19}} \pi (0.6 \times 10^{-3})^2$ $= 1.2 \times 10^{16} \text{ s}^{-1}$
6(c)	Momentum associated with each photon is $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{680 \times 10^{-9}} = 9.75 \times 10^{-28} \text{ Nm}$ Component of the momentum vertically downwards is $p_y = (9.75 \times 10^{-28}) \cos 52^\circ = 6.0 \times 10^{-28} \text{ Nm}$ For a photon that is absorbed, the change in momentum $\Delta p_{\text{absorbed}} = 6.0 \times 10^{-28} \text{ Nm}$. For a photon that is reflected, the change in momentum $\Delta p_{\text{reflected}} = 2(6.0 \times 10^{-28}) = 1.2 \times 10^{-27} \text{ Nm}$ According to Newton's second law, the force F normal to the surface as exerted by the laser light is equal to the rate of change of momentum in the vertical direction. $F = \frac{Ndp}{dt}$ $F = \frac{dN}{dt} [\Delta p_{\text{absorbed}} + \Delta p_{\text{reflected}}]$ $F = 1.2 \times 10^{16} [0.45(6.0 \times 10^{-28}) + 0.55(1.2 \times 10^{-28})]$ $= 1.12 \times 10^{-11} \text{ N}$
7(c)(i)	Random Through experiments, the radioactive decay is unpredictable and